MLOPS Assignment-2

Tracking a Pre-trained Sentiment Model with MLflow

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1. Which run had the lowest latency, and why might that be?

In the first execution, run_batch8 had the lowest latency of 0.044seconds, followed by run_batch4 (0.114 seconds) and run_batch1 (0.241 seconds).

Processing multiple sentences together reduces repeated inference-call overhead. The exact timings may vary between executions due to model warm-up and system conditions.

2. Did accuracy change between runs? Why or why not?

No. Accuracy remained 1.0 (100%) for all three batch sizes.
The same pretrained sentiment-analysis model and test sentences were used in all runs. Changing the batch size only affects how inputs are grouped for inference; it does not change the model weights or expected predictions.

3. Name two benefits of using MLflow here instead of printing results in the terminal.

1. **Reproducibility and Record-Keeping:**

MLflow saves parameters, metrics, and artifacts such as predictions.csv for future reference and inspection.

2. Easy Comparison:

MLflow displays multiple runs and their metrics side by side, making it easier to compare accuracy, confidence, and latency than reading terminal output alone

TOOLS AND TECHNOLOGIES USED:

Windows 11

Python 3.12

Visual Studio Code

MLflow 3.17.2

Pandas

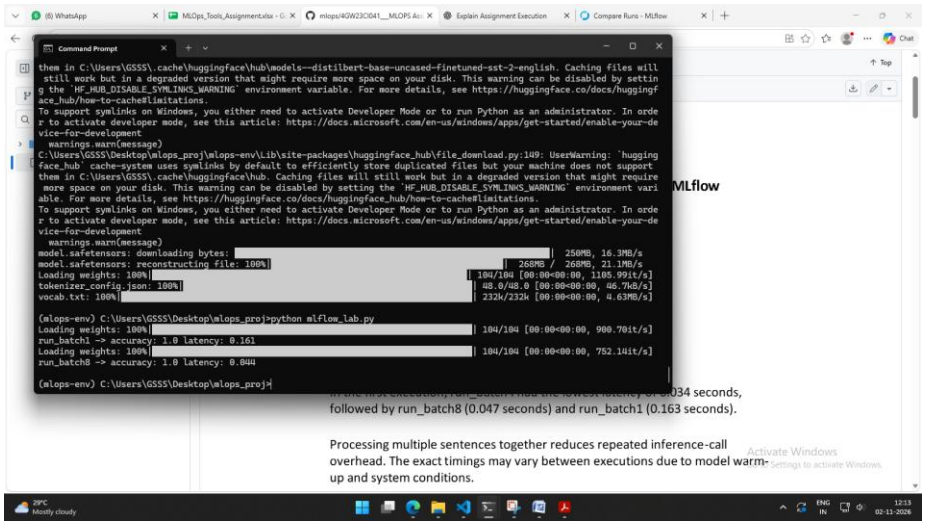
DistilBERT Pre-trained Model

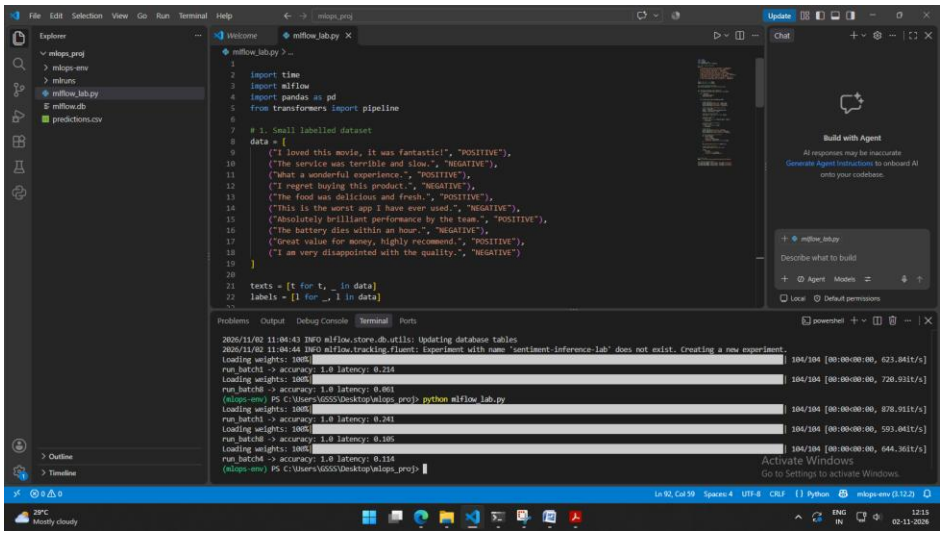
Python Virtual Environment (venv)

Command Prompt (CMD)

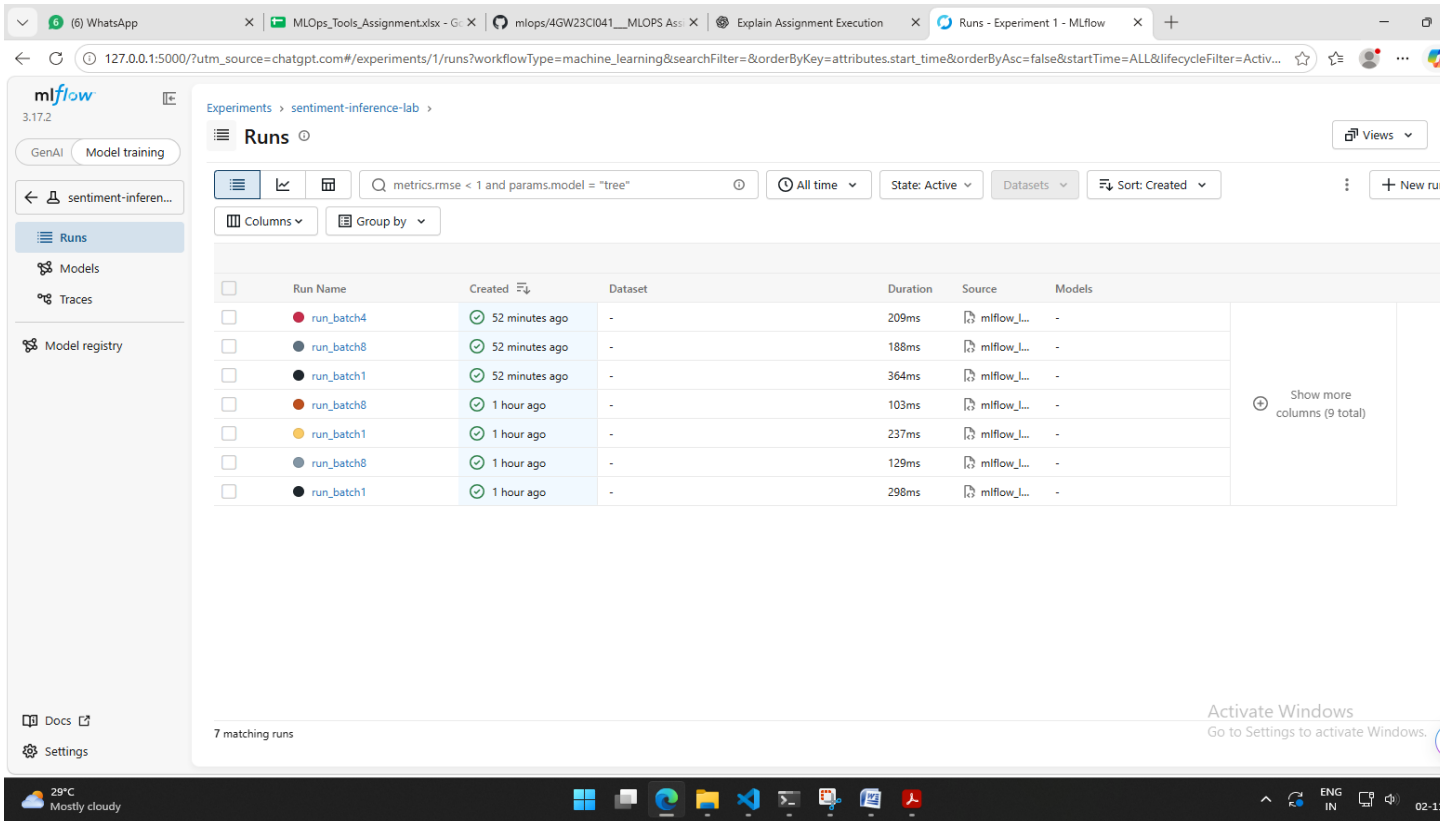
MLflow Web UI (localhost:5000)

SCREENSHOTS:-

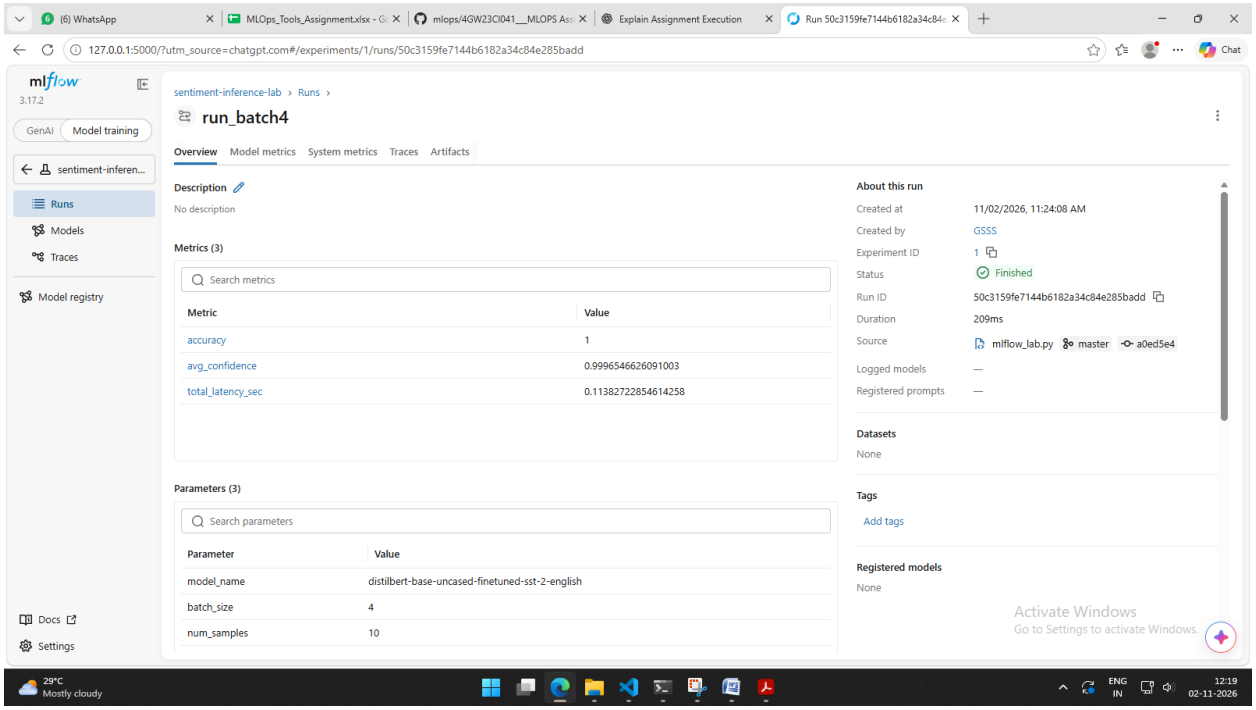



mlflow_lab.py
import time
import mlflow
import pandas as pd
from transformers import pipeline
1. Small labelled dataset
data = [
("I loved this movie, it was fantastic!", "POSITIVE"),
("The service was terrible and slow.", "NEGATIVE"),
("What a wonderful experience.", "POSITIVE"),
("I regret buying this product.", "NEGATIVE"),
("The food was delicious and fresh.", "POSITIVE"),
("This is the worst app I have ever used.", "NEGATIVE"),
("Absolutely brilliant performance by the team.", "POSITIVE"),
("The battery dies within an hour.", "NEGATIVE"),
("Great value for money, highly recommend.", "POSITIVE"),
("I am very disappointed with the quality.", "NEGATIVE")
]
texts = [t for t, _ in data]
labels = [l for _, l in data]
2026/11/02 11:04:43 INFO mlflow.store.db.utils: Updating database tables
2026/11/02 11:04:44 INFO mlflow.tracking.fluent: Experiment with name 'sentiment-inference-lab' does not exist. Creating a new experiment.
Loading weights: 100%| 104/104 [00:00<00:00, 623.84it/s]
run_batch1 -> accuracy: 1.0 latency: 0.214
Loading weights: 100%| 104/104 [00:00<00:00, 720.93it/s]
run_batch8 -> accuracy: 1.0 latency: 0.061
(mlops-env) PS C:\Users\GSSS\Desktop\mlops_proj> python mlflow_lab.py
Loading weights: 100%| 104/104 [00:00<00:00, 878.91it/s]
run_batch1 -> accuracy: 1.0 latency: 0.241
Loading weights: 100%| 104/104 [00:00<00:00, 593.04it/s]
run_batch8 -> accuracy: 1.0 latency: 0.105
Loading weights: 100%| 104/104 [00:00<00:00, 644.36it/s]
run_batch4 -> accuracy: 1.0 latency: 0.114
(mlops-env) PS C:\Users\GSSS\Desktop\mlops_proj>

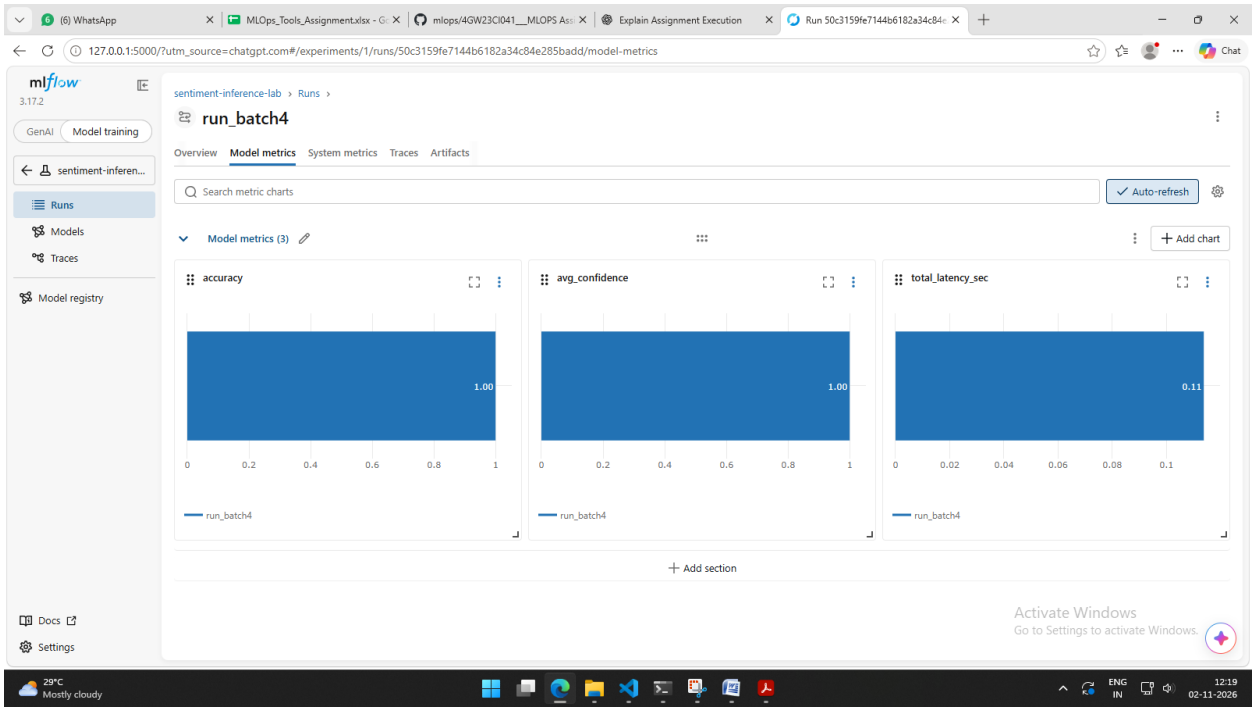
Dashboard


(6) WhatsApp
MLOps_Tools_Assignment.xlsx
mlops/4GW23CI041__MLOPS Ass
Explain Assignment Execution
Runs - Experiment 1 - MLflow
127.0.0.1:5000/?utm_source=chatgpt.com#/experiments/1/runs?workflowType=machine_learning&searchFilter=&orderByKey=attributes.start_time&orderByAsc=false&startTime=ALL&lifecycleFilter=Activ...
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
Docs
Settings
Experiments › sentiment-inference-lab ›
Runs
Views
metrics.rmse < 1 and params.model = "tree"
All time
State: Active
Datasets
Sort: Created
New ru
Columns
Group by
Run Name
Created
Dataset
Duration
Source
Models
run_batch4 52 minutes ago - 209ms mlflow_l... -
run_batch8 52 minutes ago - 188ms mlflow_l... -
run_batch1 52 minutes ago - 364ms mlflow_l... -
run_batch8 1 hour ago - 103ms mlflow_l... -
run_batch1 1 hour ago - 237ms mlflow_l... -
run_batch8 1 hour ago - 129ms mlflow_l... -
run_batch1 1 hour ago - 298ms mlflow_l... -
Show more columns (9 total)
7 matching runs
Activate Windows
Go to Settings to activate Windows.
29°C
Mostly cloudy
ENG
IN

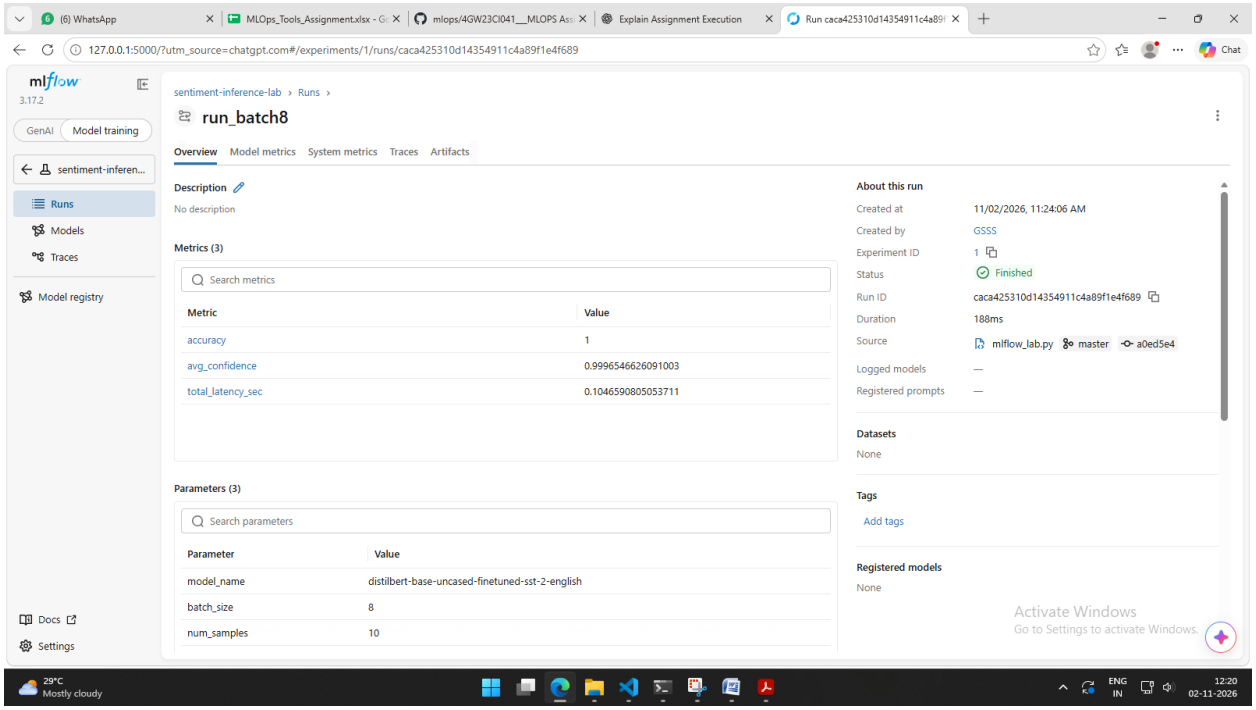
Batch Size 4:



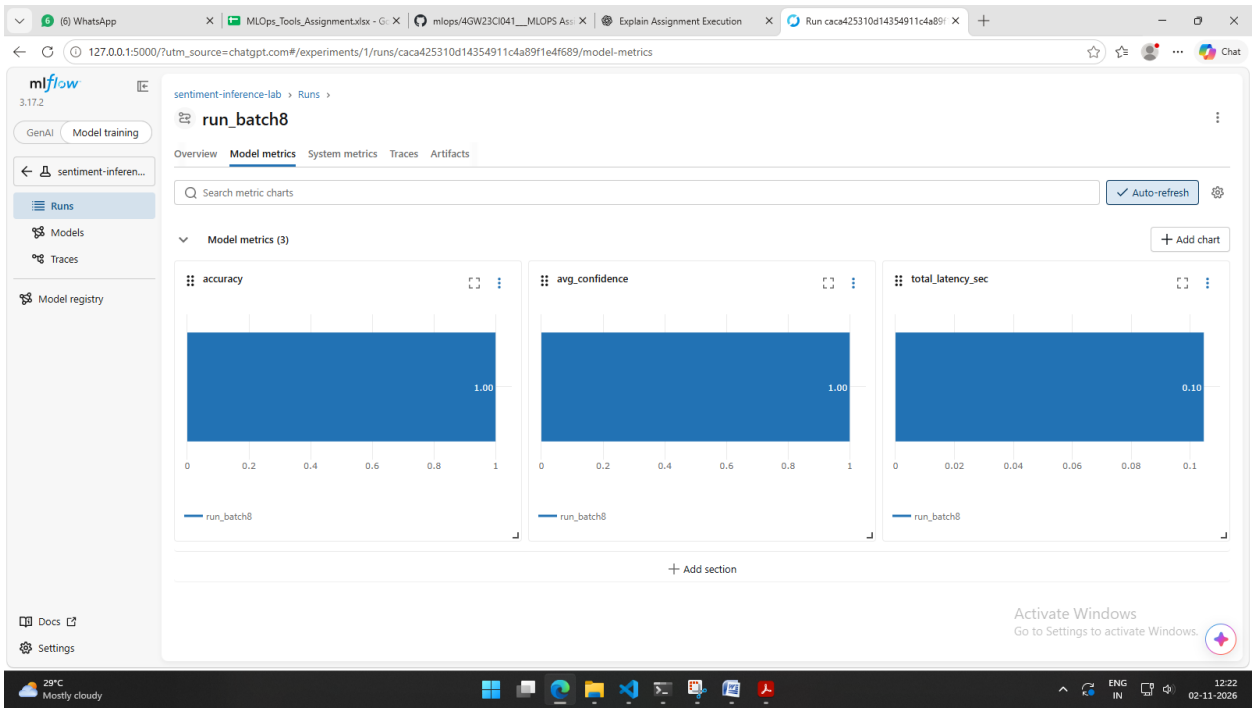
Performance Metrics and Parameters of Batch Size 4:

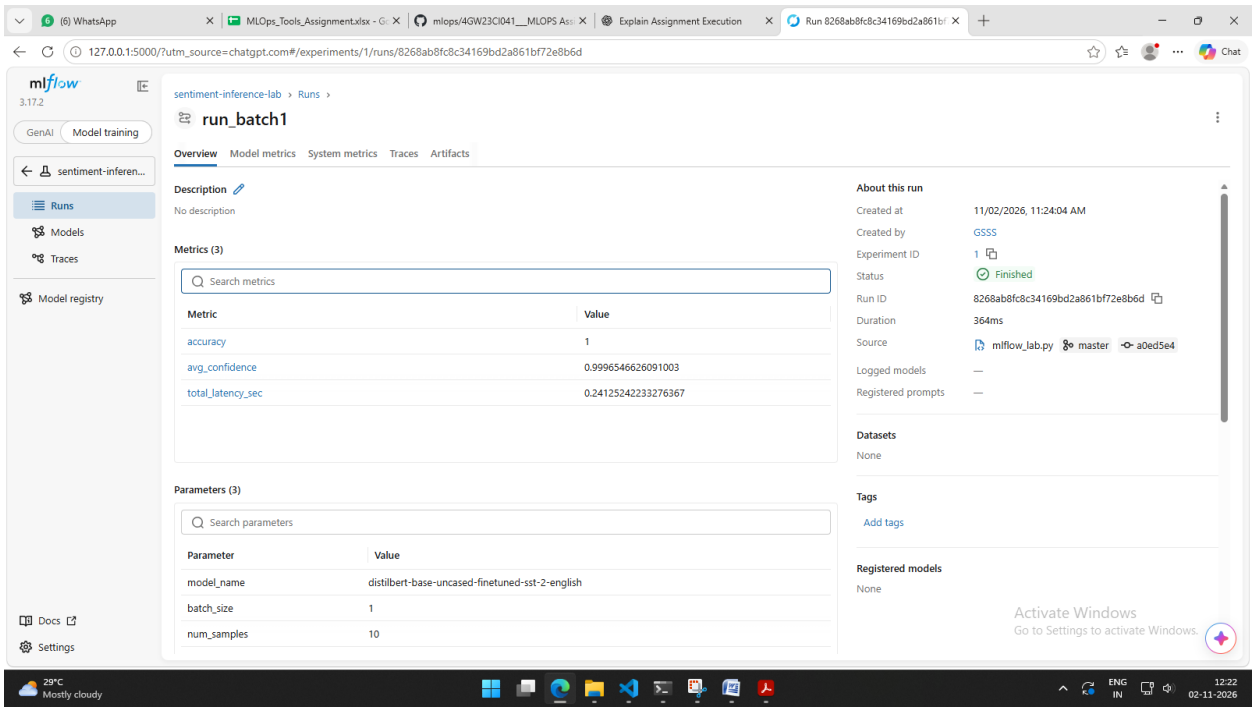


Batch Size 8:

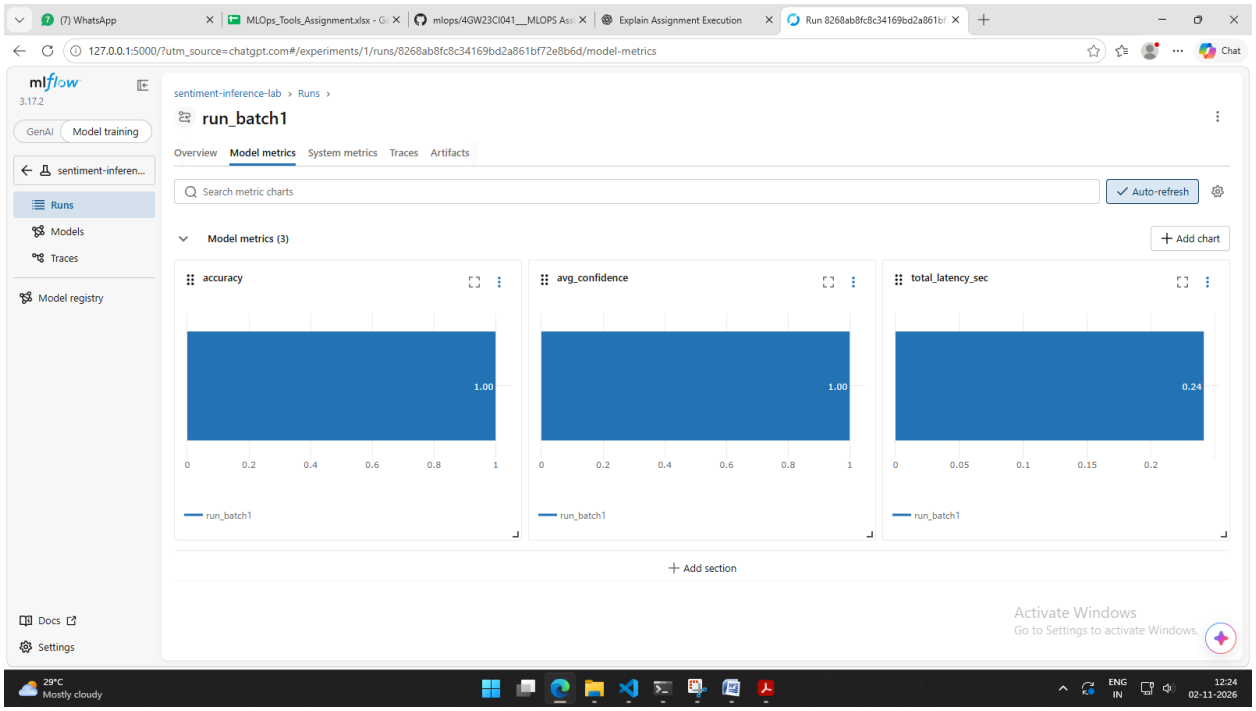


Model Performance Metrics for Batch Size 8:





Model Performance Metrics for Batch Size 1:



Comparison of Three MLflow Experiment

